

PartGen: Part-level 3D Generation and Reconstruction with Multi-View Diffusion Models

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silent-chen.github.io/PartGen

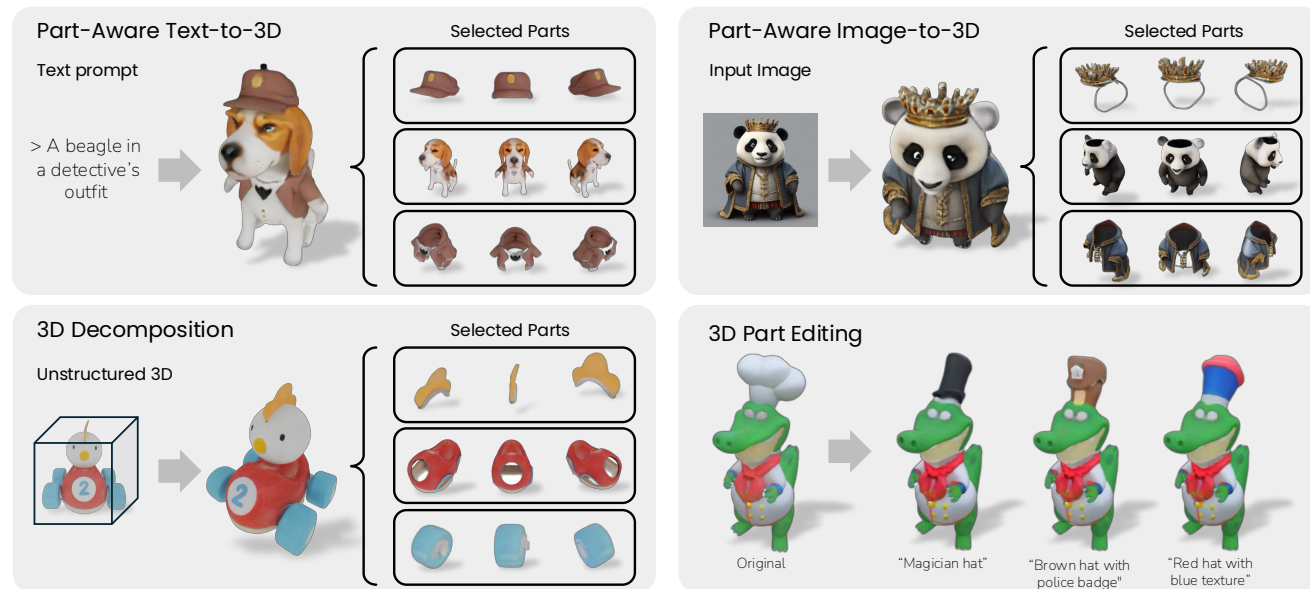


Figure 1. We introduce **PartGen**, a pipeline that generates compositional 3D objects similar to a human artist. It can start from text, an image, or an existing, unstructured 3D object. It consists of a multi-view diffusion model that identifies plausible parts automatically and another that completes and reconstructs them in 3D, accounting for their context, *i.e.*, the other parts, to ensure that they fit together correctly. Additionally, PartGen enables 3D part editing based on text instructions, enhancing flexibility and control in 3D object creation.

Abstract

Text- or image-to-3D generators and 3D scanners can now produce 3D assets with high-quality shapes and textures, but as single, fused entities lacking meaningful structure. In contrast, most applications and creative workflows require 3D assets to be composed of distinct, meaningful parts that can be independently manipulated. To bridge this gap, we introduce PartGen, a novel approach for generating, from text, images, or unstructured 3D objects, 3D objects composed of meaningful parts. Our method leverages a multi-view diffusion model to extract plausible and view-consistent part segmentations from multiple views of a

3D object, dividing it into meaningful components. A second multi-view diffusion model then processes each part individually, filling in occlusions and generating completed views, which are subsequently passed to a 3D reconstruction network. The completion process ensures that the reconstructed parts integrate cohesively by considering the context of the entire object, compensating for missing information caused by occlusions and, in extreme cases, hallucinating entirely invisible parts based on contextual cues. We evaluate PartGen on both generated and real 3D assets, demonstrating significant improvements over segmentation and part completion baselines. We also showcase downstream applications such as text-guided 3D part editing.

Work completed during Minghao Chen’s internship at Meta.

PartGen：部件级三维生成与重建：基于多视图扩散模型

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silent-chen.github.io/PartGen



图1. 我们介绍**PartGen**，这是一种能够生成类似人类艺术家创作的组合式三维对象的流程。它可以从文本、图像或现有的非结构化三维物体开始。该流程由一个能够自动识别合理部件的多视图扩散模型，以及另一个在三维空间中完成并重建这些部件的模型组成，同时考虑它们的上下文，即*i.e.* 其他部件，以确保各部件能够正确组合在一起。此外，PartGen还支持基于文本指令进行三维部件编辑，从而提升三维物体创建的灵活性和可控性。

摘要

如今，文本到三维或图像到三维生成器以及三维扫描仪已经能够生成具有高质量形状和纹理的三维资产，但这些资产通常是缺乏有意义结构的单一融合实体。相比之下，大多数应用和创作流程都要求三维资产由彼此独立且具有意义的部件组成，并能够分别进行操作。为弥合这一差距，我们提出了**PartGen**，这是一种新颖的方法，能够从文本、图像或非结构化三维物体生成由有意义部件组成的三维物体。我们的方法利用多视图扩散模型，从某个物体的多个视图中提取合理且视图一致的部件分割

Minghao Chen 在 Meta 实习期间完成的工作。

1. Introduction

It is now possible to obtain high-quality 3D objects from text or images via 3D generation [12, 14, 18, 47, 52, 54, 73, 79] and 3D scanning [15, 59, 85]. However, the objects obtained in this manner are usually monolithic and *unstructured*, represented by a single neural field, mixture of Gaussians, or mesh. This is insufficient in a professional setting, where the *structure* of an asset is of paramount importance. For example, 3D artists often use parts to facilitate reusing, editing, and animating 3D objects.

In this paper, we thus consider the problem of obtaining *structured* 3D objects that comprise several meaningful *parts*, akin to the models produced by 3D artists. For instance, an artist may wish to decompose the 3D model of a person into its clothes and accessories, as well as various anatomical features like hair, eyes, teeth, and limbs. When a 3D object is generated or scanned, however, its parts are usually ‘fused’ together. This is undesirable because the parts may carry different semantics, materials, or motion. Parts may also be physically detachable and interchangeable, and an artist may wish to remove, replace, or edit them independently. In video games, parts often need to be reconfigured dynamically, *e.g.*, to represent a character picking up a weapon or changing clothes. Parts are also important for 3D understanding in robotics, embodied AI, and *spatial intelligence* [44, 49].

Inspired by these requirements, we introduce **PartGen**, a method to upgrade existing 3D generation pipelines from producing unstructured 3D objects to generating compositions of meaningful 3D parts (Fig. 1). To do so, we address two key questions: (1) how to automatically *segment* a 3D object into parts, and (2) how to extract high-quality, *complete* 3D parts even when these are only partially—or not at all—visible from the exterior of the 3D object.

Part segmentation and completion are highly ambiguous tasks. Since different artists may find it useful to decompose the same object in different ways, there is no ‘gold-standard’ segmentation for any given 3D object. Hence, the goal is to model the distribution of plausible part decompositions rather than a single one. Furthermore, current 3D reconstruction and generation methods only model an object’s visible outer surface, omitting internal details. Therefore, decomposing an object into parts often requires completing or even hallucinating them entirely.

To model this ambiguity, we base part segmentation and completion on generative models. We note that most state-of-the-art 3D generation pipelines [12, 14, 18, 36, 47, 52, 54, 73, 79] start by generating several consistent 2D views of the object, and then apply a 3D reconstruction network to those images to recover the 3D object. We build upon this two-stage scheme and address part segmentation and completion before 3D reconstruction is carried out.

First, we cast part segmentation as a *stochastic multi-*

view-consistent coloring problem, fine-tuning a multi-view image generator to produce several color-coded segmentation maps of the given 3D object. We do not assume any explicit or deterministic taxonomy of parts. Instead, we train the model on a large collection of artist-created data, capturing how 3D artists *typically* decompose objects into parts. This approach has two key benefits. First, it leverages an image generator that is already trained to be view-consistent. Second, it reduces obtaining multiple plausible segmentations of the object to drawing multiple samples from the model. In practice, we show that this approach better captures the artists’ intent compared to fine-tuning a model like SAM [33] or SAM2 [67] for the same task.

Second, we fine-tune another multi-view image generator to *complete* the segmented parts while *accounting for the context* of the object as a whole. In this manner, parts which are only partially visible, or even invisible, in the original input views, are revealed in full to the 3D reconstruction network. This has the benefit of removing most of the ambiguity in the reconstruction task, which is important because the 3D reconstructor network is deterministic and cannot handle ambiguity well. Furthermore, because completion is contextual, the resulting parts fit together well, forming a coherent 3D object despite the fact that the parts are reconstructed independently.

We show that PartGen can be applied to different input modalities: starting from text, an image, or a real-world 3D scan, PartGen can generate 3D assets with meaningful parts. We assess our method empirically on a large collection of 3D assets produced by 3D artists or scanned, both quantitatively and qualitatively. We also demonstrate that PartGen can be easily extended to the task of 3D part editing.

2. Related Work

3D generation from text and images. The problem of generating 3D assets from text or images has been thoroughly studied in the literature. Some authors have built generators from scratch. For instance, CodeNeRF [28] learns a latent code for NeRF in a Variational Autoencoder fashion, while Shap-E [29] and 3DGen [21] do so using latent diffusion. PC² [51] and Point-E [58] diffuse a point cloud, and MosaicSDF [91] uses a semi-explicit SDF-based representation. However, 3D training data is scarce, which makes it difficult to train text-based 3D generators directly.

DreamFusion [62] demonstrated for the first time that 3D assets can be extracted from text-to-image (T2I) diffusion models with the *Score Distillation Sampling* (SDS) loss. Variants of DreamFusion explore representations like hash grids [38, 63], meshes [38], and 3D Gaussians (3DGS) [9, 75, 93], tweaks to the SDS loss [26, 81, 83, 101], conditioning on an input image [50, 63, 74, 76, 95], regularizing normals or depth [65, 71, 74], and extensions of SDS to 3D editing tasks [7, 60, 69, 72, 102, 103].

1. 引言

现在可以通过3D生成 [12, 14, 18, 47, 52, 54,73, 79] 和 3D扫描 [15, 59, 85]从文本或图像中获取高质量的三维物体。然而，以这种方式获取的物体通常是单一整体且无结-构的，由单个神经场、高斯混合模型或网格表示。这在专业环境中是不够的，因为资产的结构至关重要。例如，3D艺术家经常使用部件，以便于重复使用、编辑和制作三维物体的动画。

因此，本文研究了获取由多个有意义的结构化三维物体部件组成的三维物体这一问题，这类模型类似于3D艺术家制作的模型。例如，艺术家可能希望将人物的三维模型分解为衣物和配饰，以及头发、眼睛、牙齿和四肢等各种解剖特征。然而，当三维物体被生成或扫描时，其各个部件通常会被“融合”在一起。这并不理想，因为这些部件可能具有不同的语义、材质或运动方式。部件还可能在物理上可拆卸且可互换，艺术家可能希望独立地移除、替换或编辑这些部件。在电子游戏中，部件通常需要动态重新配置，例如例如，用于表示角色拾起武器或更换衣物。对于机器人学、具身人工智能和空间智能 [44, 49]而言，部件对于三维理解也十分重要。

受这些需求启发，我们提出了**PartGen**，这是一种将现有三维生成流程从生成非结构化三维物体升级为生成由有意义的三维部件组成的组合体的方法（图1）。为此，我们需要解决两个关键问题：（1）如何自动分割三维物体中的部件；以及（2）即使这些部件只能从三维物体外部部分可见，或完全不可见，如何提取高质量的、完整的三维部件。

部件分割和补全是高度模糊的任务。由于不同的艺术家可能会发现，以不同方式对同一个物体进行分解更为有用，因此对于任何给定的三维物体，都不存在“黄金标准”的分割结果。因此，目标是对合理的部件分解方式的分布进行建模，而不是只建模一种分解方式。此外，当前的三维重建和生成方法只能对物体可见的外部表面进行建模，而忽略内部细节。因此，将物体分解为各个部件通常需要对这些部件进行补全，甚至完全凭空构想出来。

为了对这种模糊性进行建模，我们基于生成模型开展部件分割和补全。我们注意到，大多数最先进的三维生成流程 [12, 14, 18, 36, 47, 52,54, 73, 79] 首先生成物体的多个一致二维视图，然后将三维重建网络应用于这些图像，以恢复三维物体。我们在这一两阶段方案的基础上，在进行三维重建之前处理部件分割和补全问题。

首先，我们将部件分割视为一个随机多

视图一致的着色问题，通过微调多视图图像生成器，为给定的三维物体生成多个颜色编码的分割图。我们不假设任何显式或确定性的部件分类体系。相反，我们在大量由艺术家创建的数据集上训练模型，捕捉3D艺术家通常如何将物体分解为各个部件。这种方法有两个主要优势。首先，它利用了一个已经过训练、能够保持视图一致的图像生成器。其次，它将获取物体的多个合理分割结果，简化为从模型中绘制多个样本。在实践中，我们展示了，与针对同一任务微调类似 SAM [33] 或 SAM2 [67] 的模型相比，这种方法能更好地捕捉艺术家的意图。

其次，我们对另一个多视图图像生成器进行微调，以补全分割部件，同时考虑物体整体的上下文。通过这种方式，在原始输入视图中仅部分可见甚至完全不可见的部件，能够完整地呈现在三维重建器网络中。这样做的好处是消除了重建任务中的大部分歧义，这一点很重要，因为三维重建器网络是确定性的，难以妥善处理歧义。此外，由于补全过程具有上下文关联性，最终得到的部件能够良好地拼接在一起，即使这些部件是独立重建的，也能形成连贯的三维物体。

我们展示了 PartGen 可应用于不同的输入模态：从文本、图像或真实世界的三维扫描开始，PartGen 能够生成包含有意义部件的三维资产。我们在由3D艺术家制作或扫描得到的大规模三维资产集合上，从定量和定性两方面对我们的方法进行了实证评估。我们还证明，PartGen 可以轻松扩展到三维部件编辑任务。

2. 相关工作

从文本和图像生成三维。 从文本或图像生成三维资产的问题在文献中已得到充分研究。一些研究者从头开始构建生成器。例如，CodeNeRF [28]以变分自编码器的方式学习NeRF的潜在代码，而Shap-E [29] 和 3DGen [21] 则使用潜在扩散模型实现这一点。PC² [51]和Point-E [58]对点云进行扩散，MosaicSDF [91] 则使用基于SDF的表示。然而，三维训练数据十分稀缺，因此难以直接训练基于文本的三维生成器。

DreamFusion [62] 首次证明，可以通过带有分数蒸馏采样（SDS）损失的文本到图像（T2I）扩散模型提取三维资产。DreamFusion 的变体探索了哈希网格 [38, 63], 网格 [38], 和三维高斯（3DGS）[9, 75, 93], 等表示形式、对 SDS 损失的调整 [26, 81, 83, 101]、以输入图像为条件 [50, 63, 74, 76, 95]、对法线或深度进行正则化 [65, 71, 74], , 以及将 SDS 扩展到三维编辑任务 [7, 60, 69, 72, 102, 103]。

Other works focus on improving the 3D awareness of the T2I model, simplifying the extraction of a 3D output and eschewing the need for slow SDS optimization. Inspired by 3DIM [84], Zero-1-to-3 [43] fine-tunes the 2D generator to output novel views of the object. Two-stage approaches [6, 10, 18, 22–24, 41, 45, 46, 52, 71, 77, 82, 86, 88–90] take the output of a text- or image-to-multi-view model that generates multiple views of the object and reconstruct the latter using multi-view reconstruction methods like NeRF [55] or 3DGS [30]. Other approaches reduce the number of input views generated and learn a fast feed-forward network for 3D reconstruction. Perhaps the most notable example is Instant3D [36], based on the *Large Reconstruction Model* (LRM) [25]. Recently, there has been a focus on 3D compositional generation [11, 37, 61, 104]. D3LL [17] learns 3D object composition through distillation from a 2D T2I generator. ComboVerse [8] starts from a single image, but mostly at the level of different objects instead of their parts, performing single-view inpainting and reconstruction, and using SDS for composition.

3D segmentation. Our work decomposes a given 3D object into parts. Several works have considered segmenting 3D objects or scenes represented in an unstructured manner, often as neural fields or 3D Gaussian mixtures. Semantic-NeRF [98] was the first to fuse 2D semantic segmentation maps in 3D with neural fields. DFF [34] and N3F [80] propose mapping 2D features to 3D fields, allowing for supervised and unsupervised segmentation. LERF [31] extends this concept to language-aware features like CLIP [66]. Contrastive Lift [2] considers instance segmentation, fusing information from several independently segmented views using a contrastive formulation. GARField [32] and OmniSeg3D [94] consider that concepts exist at different levels of scale, which they identify with the help of SAM [33]. LangSplat [64] leverages both CLIP and SAM, creating distinct 3D language fields to model each SAM scale explicitly, while N2F2 [3] automates binding the correct scale to each concept. Neural Part Priors [4] completes and decomposes 3D scans with learned part priors in a test-time optimization manner. Finally, Uni3D [99] learns a ‘foundation’ model for 3D point clouds, enabling zero-shot segmentation.

Primitive-based representations. Some authors propose representing 3D objects as a mixture of primitives [96], which can be seen as related to parts, although they are usually non-semantic. For example, SIF [19] represents an occupancy function as a mixture of 3D Gaussians. LDIF [20] uses the Gaussians to window local occupancy functions implemented as neural fields [53]. Neural Template [27] and SPAGHETTI [1] learn to decompose shapes in a similar manner using an auto-decoding setup. SALAD [35] uses SPAGHETTI as the latent representation for a diffusion-based generator. PartNeRF [78] is conceptually similar

but builds a mixture of NeRFs. NeuForm [39] and Diff-Facto [57] learn representations that afford part-based control. DBW [56] decomposes real-world scenes with textured superquadric primitives.

Semantic Part-Based Representations. Others have considered 3D parts that are semantic. PartSLIP [42] and Part-SLIP++ [100] use vision-language models to segment objects into parts using point clouds as the representation. Part123 [40] is conceptually similar to Contrastive Lift [2], but applied to objects rather than scenes, and to the output of a monocular reconstruction network instead of NeRF.

In this paper, we address a problem different from the ones above. We generate compositional 3D objects from various modalities using multi-view diffusion models for segmentation and completion. Parts are meaningfully segmented, fully reconstructed, and correctly assembled. We handle the ambiguity of these tasks in a generative way.

3. Method

This section introduces PartGen, our framework for generating 3D objects composed of several 3D parts. Each part is a distinct, human-interpretable, self-contained and *complete* element. PartGen can take different modalities as input (text prompts, image prompts, or 3D assets) and performs part segmentation and completion by repurposing a powerful multi-view diffusion model for these two tasks. An overview of PartGen is given in Fig. 2.

The rest of the section is organized as follows. In Sec. 3.1, we introduce the necessary background on multi-view diffusion and briefly describe how PartGen can be applied to text, image, or 3D model inputs. Then, in Secs. 3.2 to 3.4, we describe how PartGen automatically segments, completes, and reconstructs meaningful parts in 3D. In Sec. 3.5, we describe the dataset used to train PartGen.

3.1. Background on 3D generation

First, we provide essential background on multi-view diffusion models for 3D generation [36, 71, 73]. These methods typically adopt a two-stage approach to 3D generation.

In the first stage, given a prompt y , an image generator Φ outputs several 2D views of the object from different vantage points. Depending on the nature of y , the network Φ is either a text-to-image (T2I) model [36, 71] or an image-to-image (I2I) model [70, 82]. Here, we assume that the network outputs four views from the four cardinal directions around the object, each with a resolution of $H \times W$; these are arranged in a 2×2 grid, forming a single ‘multi-view’ image $I \in \mathbb{R}^{3 \times 2H \times 2W}$. This model thus provides a probabilistic mapping $I \sim p(I \mid \Phi, y)$. In addition to text and images, the input y can also be an existing 3D model. In this case, I is obtained by *rendering* the 3D model from the four cardinal views.

其他工作致力于提升 T2I 模型的三维感知能力，简化三维输出的提取过程，并避免耗时的 SDS 优化。受 3DIM [84], 启发，Zero-1-to-3 [43] 对二维生成器进行微调，使其输出物体的新视角。两阶段方法 [6, 10, 18, 22–24, 41, 45, 46, 52, 71, 77, 82, 86, 88–90] 以文本到多视图或图像到多视图模型的输出为基础，生成物体的多个视图，并使用 NeRF [55] 或 3DGS [30]等多视图重建方法对后者进行重建。其他方法减少生成的输入视图数量，并学习用于三维重建的快速前馈网络。其中最著名的例子或许是基于大型重建模型（LRM）的 Instant3D [36], [25]。近年来，研究重点转向三维对象组合生成 [11, 37, 61, 104]。D3LL [17] 通过从二维 T2I 生成器进行蒸馏，学习三维对象组合。ComboVerse [8] 从单张图像开始，但主要在不同物体层面而非其部件层面进行处理，执行单视图修复和重建，并使用 SDS 进行组合。

三维分割。 我们将给定的三维物体分解为各个部件。已有多项工作研究了对以非结构化方式表示的三维物体或场景进行分割，这类表示通常采用神经场或三维高斯混合。Semantic-NeRF [98] 首次将二维语义分割图与神经场融合到三维中。DFF [34] 和 N3F [80] 提出将二维特征映射到三维场，从而实现有监督分割和无监督分割。LERF [31] 将这一概念扩展到类似 CLIP [66]的语言感知特征。Contrastive Lift [2]研究实例分割，通过对比学习形式融合来自多个独立完成分割的视图的信息。GARField [32] 和 OmniSeg3D [94] 认为，不同概念存在于不同的尺度层级，并借助 SAM [33]识别这些尺度。LangSplat [64] 同时利用 CLIP 和 SAM，创建彼此独立的三维语言场，以显式建模 SAM 的每个尺度，而 N2F2 [3] 则自动将每个概念绑定到正确的尺度。神经部件先验 [4]以测试时优化的方式，利用学习到的部件先验补全并分解三维扫描。最后，Uni3D [99] 学习用于三维点云的“基础”模型，从而实现零样本分割。

基于基元的表示。 一些作者提出将三维物体表示为基元的混合体 [96],, 这些基元可以看作与部件相关，尽管它们通常不具备语义。例如，SIF [19] 将占据函数表示为三维高斯的混合体。LDIF [20]使用高斯函数对由神经场 [53]实现的局部占据函数进行窗口化。Neural Template [27]和 SPAGHETTI [1]采用类似的方式，通过自动解码设置学习分解形状。SALAD [35]将 SPAGHETTI 作为基于扩散的生成器的潜在表示。PartNeRF [78] 在概念上类似

但它构建的是 NeRF 的混合体。NeuForm [39] 和 Diff-Facto [57] 学习能够实现基于部件控制的表示。DBW [56] 使用带纹理的超二次曲面基元分解真实世界场景。

语义部件表示。 其他研究者也考虑过具有语义的三维部件。PartSLIP [42]和 Part-SLIP++ [100] 使用视觉语言模型，以点云作为表示将物体分割成各个部件。Part123 [40] 在概念上类似于 Contrastive Lift [2],, 但其应用对象是物体而非场景，且作用于单目重建网络的输出，而非 NeRF。

在本文中，我们解决的问题不同于上述问题。我们使用多视图扩散模型进行分割和补全，从各种模态生成组合式三维对象。各部件会被有意义地分割、完整重建并正确组装。我们以生成式方式处理这些任务中的歧义。

3. 方法

本节介绍 PartGen，这是我们的框架，用于生成由多个三维部件组成的三维对象。每个部件都是独立的、可由人理解的、自包含且完整的元素。PartGen 可以接收不同模态的输入（文本提示、图像提示或三维资产），并通过重新利用强大的多视图扩散模型来执行部件分割和补全任务。PartGen 的概览如图2所示。

本节其余部分的组织如下。在第 3.1 节中，我们介绍多视图扩散的必要背景，并简要说明如何将 PartGen 应用于文本、图像或 3D模型输入。随后，在第 3.2 至 3.4 节中，我们介绍 PartGen 如何自动对三维场景中的有意义的部件进行分割、补全和重建。在第 3.5 节中，我们介绍用于训练 PartGen 的数据集。

3.1. 三维生成背景

首先，我们介绍用于 3D生成的多视图扩散模型的基本背景知识 [36, 71, 73]。这些方法通常采用两阶段的 3D生成方案。

在第一阶段，给定提示词 y ，图像生成器！从不同的观察视角生成物体的多个二维视图。根据 y 的性质，网络！要么是文本到图像（T2I）模型 [36, 71]，要么是图像到图像（I2I）模型 [70, 82]。在此，我们假设网络从物体周围的四个基本方向输出四个视图，每个视图的分辨率为 $H \times W$ ；这些视图排列成一个 2×2 网格，形成单张“多视图”图像 $I \in \mathbb{R}^{3 \times 2H \times 2W}$ 。因此，该模型提供了一个概率映射 $I \sim p(I \mid \Phi, y)$ 。除了文本和图像之外，输入 y 还可以是现有的3D模型。在这种情况下， I 通过从四个基本方向对3D模型进行渲染获得。

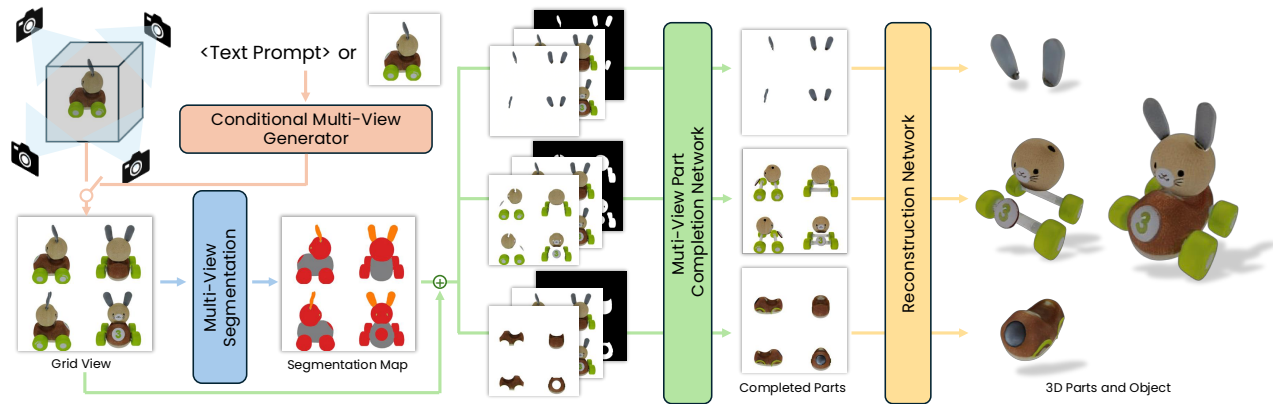


Figure 2. **Overview of PartGen.** Our method begins with text, single images, or existing 3D objects to obtain an initial grid view of the object. This view is then processed by a diffusion-based segmentation network to achieve multi-view consistent part segmentation. Next, the segmented parts, along with contextual information, are input into a multi-view part completion network to generate a fully completed view of each part. Finally, a pre-trained reconstruction model generates the 3D parts.

The 2D views I are subsequently passed to a Reconstruction Model (RM) [36, 73, 87], denoted as Ψ , *i.e.*, a neural network that reconstructs the 3D object $\mathbf{L}$ in both shape and appearance. The generator Φ is obtained by fine-tuning an image generation model pre-trained on Internet-scale 2D data, which is a significant advantage compared to learning a 3D generator directly, as 3D training data is scarce.

There are many possible choices for the base image generation and 3D reconstruction models. In the experiments, we follow AssetGen [73] and obtain Φ by fine-tuning a pre-trained text-to-image diffusion model with an architecture similar to Emu [13], an image generator based on latent diffusion using 8 latent dimensions. Details of the fine-tuning strategy are given in Sec. 4.4 and the sup. mat. For the RM Ψ , we use LightplaneLRM [5], trained on our dataset.

3.2. Multi-view part segmentation

The first contribution of our paper is a method for decomposing a 3D object into its constituent parts. Inspired by multi-view diffusion approaches, we frame this decomposition problem as a *multi-view segmentation* task instead of direct 3D segmentation. Specifically, the goal is to map I to a collection of 2D masks $M^1, \dots, M^S \in \{0, 1\}^{2H \times 2W}$, one for each visible part of the object. Notably, both the image I and the masks M_i are multi-view grids.

Addressing 3D object segmentation through the lens of multi-view diffusion offers several advantages. First, it allows us to repurpose existing multi-view models Φ , which, as described in Sec. 3.1, are already pre-trained to produce multi-view consistent generations in the RGB domain. Second, it integrates easily with established multi-view frameworks. Third, decomposing an object into parts is an inherently non-deterministic, ambiguous task as it depends on the desired granularity, application requirements, and artistic intent. By learning this task with probabilistic diffusion models, we can capture this ambiguity effectively.

We train the model using a dataset of artist-created 3D objects (Sec. 3.5), where each object $\mathbf{L}$ is annotated with a possible decomposition into 3D parts, $\mathbf{L} = (\mathbf{S}^1, \dots, \mathbf{S}^S)$. The input to the model is a multi-view image I , and the output is a set of multi-view part masks $M^1, M^2, \dots, M^S$ corresponding to parts $\mathbf{S}^1, \dots, \mathbf{S}^S$. To fine-tune the multi-view image generators Φ for this task, we quantize the RGB space into Q different (and fixed) colors $c_1, \dots, c_Q \in [0, 1]^3$. For each training sample $\mathbf{L} = (\mathbf{S}^k)_{k=1}^S$, we assign colors to the parts, mapping part $\mathbf{S}^k$ to color c_{π_k} , where π is a random permutation on $\{1, \dots, Q\}$ (we assume that $Q \geq S$). Given this mapping, we render the segmentation map as a multi-view RGB image $C \in [0, 1]^{3 \times 2H \times 2W}$ (Fig. 4). Then, we obtain Φ_{seg} by fine-tuning Φ to: (1) take as conditioning the multi-view image I , and (2) generate the color-coded multi-view segmentation map C , hence sampling $C \sim p(C | \Phi_{\text{seg}}, I)$.

This approach can produce alternative segmentations by simply re-running Φ_{seg} , which is stochastic. It further exploits the fact that Φ_{seg} is stochastic to discount the specific ‘naming’ or coloring of the parts, which is arbitrary. Naming is a technical issue in instance segmentation that usually requires ad-hoc solutions, but here it is implicitly addressed by the stochastic nature of the model.

To extract the segments at test time, we sample the color-coded segmentation map C and simply quantize it based on the reference colors $c_1, \dots, c_Q$, discarding parts that contain only a few pixels.

Implementation details. The network Φ_{seg} has the same architecture as the network Φ with some changes to allow conditioning on the multi-view image I : we encode it into latent space with the VAE and stack it with the noised latent as the input to the diffusion network.

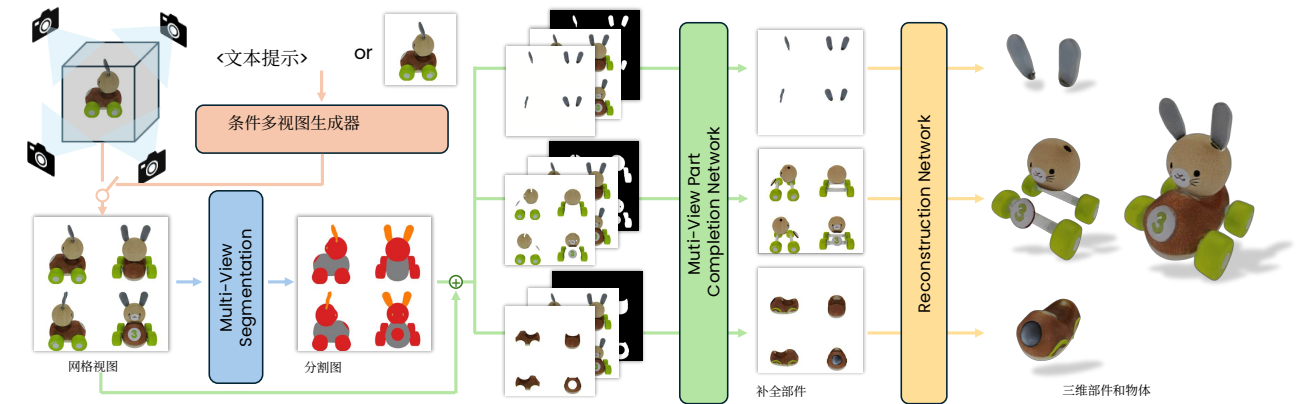


图 2. **PartGen 概览。** 我们的方法首先以文本、单张图像或现有的三维物体为输入，获取物体的初始网格视图。随后，通过基于扩散的分割网络处理该视图，实现多视图一致的部件分割。接下来，将分割部件及上下文信息输入多视图部件补全网络，为每个部件生成完整的视图。最后，预训练的重建模型生成3D部件。

二维视图 I 随后被传递给一个重建模型 (RM) [36, 73, 87], 记作 Ψ , 即, 一种能够在形状和外观两方面重建三维物体的神经网络 $\mathbf{L}$ 。生成器 Φ 通过对一个在互联网规模二维数据上预训练的图像生成模型进行微调获得。与直接学习三维生成器相比, 这是一个显著优势, 因为三维训练数据十分稀缺。

用于基础图像生成和三维重建模型的选择有很多。在实验中, 我们遵循 AssetGen [73] 并通过以下方式获得! : 对采用类似 Emu 架构的预训练文本到图像扩散模型进行微调 [13], 这是一个基于潜在扩散、使用 8 个潜在维度的图像生成器。微调策略的详细信息见第 4.4 节及补充材料。对于 $\text{RM}\Psi$, 我们使用在自身数据集上训练的 LightplaneLRM [5]。

3.2. 多视图部件分割

本文的第一个贡献是提出一种将三维物体分解为其组成部件的方法。受多视图扩散方法的启发, 我们将这一分解问题定义为多视图分割任务, 而不是直接进行三维分割。具体而言, 目标是将 I 映射为一组二维掩码

$M^1 \dots M^S \in \{0, 1\}^{2H \times 2W}$, 每个可见部件对应一个掩码。值得注意的是, 图像 I 和掩码 M_i 都是多视图网格。

从多视图扩散的角度处理三维物体分割具有多项优势。首先, 它使我们能够重新利用现有的多视图模型!, 正如第 3.1 节所述, 这些模型已经过预训练, 能够在 RGB 域中生成多视图一致的结果。其次, 它可以轻松地与现有的多视图框架集成。第三, 将物体分解为部件本质上是一项非确定性的模糊任务, 因为它取决于所需的粒度、应用需求和艺术意图。通过使用概率扩散模型学习这一任务, 我们可以有效地捕捉这种模糊性。

我们使用一个由艺术家创作的三维物体数据集 (第 3.5 节) 训练模型, 其中每个物体 $\mathbf{L}$ 都标注了一种可能的三维部件分解方式, $\mathbf{L} = (\mathbf{S}^1, \dots, \mathbf{S}^S)$ 。模型的输入是多视图图像 I , 输出是一组三维部件的多视图部件掩码 $M^1 M^2 \dots M^S$ 、, 分别对应于部件 $\mathbf{S}^1 \dots \mathbf{S}^S$ 、。为了针对这一任务微调多视图图像生成器!, 我们将 RGB 空间量化为 Q 种不同且固定的颜色 $c_1 \dots c_Q \in [0, 1]^3$ 。对于每个训练样本 $\mathbf{L} = (\mathbf{S}^k)_{k=1}^S$, 我们为各个部件分配颜色, 将部件 $\mathbf{S}^k$ 映射为颜色 c_{π_k} , 其中 π 是 $\{1, \dots, Q\}$ 上的一个随机排列 (我们假设 $Q \geq S$)。给定这一映射后, 我们将分割图渲染为多视图 RGB 图像 $C \in [0, 1]^{3 \times 2H \times 2W}$ (图 4)。随后, 我们通过微调! 获得! seg , 使其: (1) 以多视图图像 I 作为条件输入; (2) 生成颜色编码的多视图分割图 C , 从而采样 $C \sim p(C | \Phi_{\text{seg}}, I)$ 。

只需重新运行! seg (该过程具有随机性), 这种方法即可生成替代分割结果。此外, 它还利用了! seg 具有随机性的事实, 从而弱化部件具体“命名”或着色的影响, 因为这些命名或着色本身是任意的。在实例分割中, 命名是一个通常需要临时解决方案的技术问题, 但在这里, 模型的随机性隐式地解决了这一问题。

为了在测试时提取分割区域, 我们对颜色编码的分割图 C 进行采样, 然后根据参考颜色 $c_1, \dots, c_Q$ 对其进行简单量化, 舍弃仅包含少量像素的部分。

实现细节。 该网络! seg 与网络! 具有相同的架构, 但进行了部分修改, 以便能够以多视图图像 I 作为条件: 我们使用变分自编码器将其编码到潜在空间中, 并将其与加噪潜变量堆叠起来, 作为扩散网络的输入。

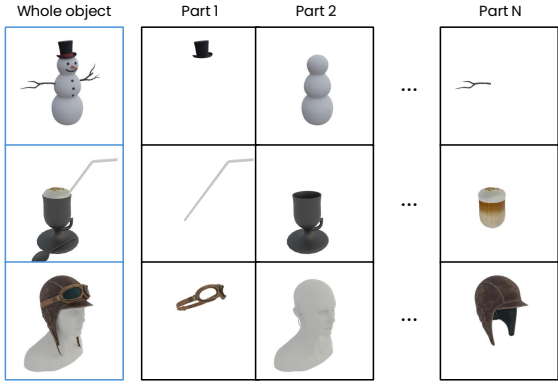


Figure 3. **Training data.** We obtain a dataset of 3D objects decomposed into parts from artist-created assets. These come ‘naturally’ segmented into parts according to the artist’s design.

3.3. Contextual part completion

Our method so far has produced a multi-view image I of the 3D object along with 2D segments $M^1, M^2, \dots, M^S$. What remains is to convert these into *full* 3D part reconstructions. Given a mask M , we could submit the masked image $I \odot M$ to the RM Ψ to obtain a 3D reconstruction of the part, *i.e.*, $\hat{S} = \Psi(I \odot M)$. However, some parts can be heavily occluded by other parts and, in extreme cases, may be entirely invisible. While we could train the RM to handle such occlusions, in practice, this approach does not work well because part completion is inherently a stochastic problem, whereas the RM is deterministic.

To solve this problem, we repurpose yet again the multi-view generator Φ , this time to perform part completion. The latter model is able to generate a full object from text or a single image, so, properly fine-tuned, it should be able to hallucinate any missing portion of a part.

Formally, we fine-tune Φ into a completion model Φ_{comp} which maps the masked multi-view part image $I \odot M$ to the completed multi-view part image $\hat{J} \sim p(\hat{J} \mid \Phi_{\text{comp}}, I \odot M)$. However, we note that sometimes parts are barely visible, so the masked image $I \odot M$ provides very little information. Because we need the generated part to fit well with the other parts and the whole object, we provide the model with the unmasked image I for *context*. Thus, we condition $p(\hat{J} \mid I \odot M, I, M)$ on the masked image $I \odot M$, the unmasked image I , and the mask M . The importance of the context I increases with the extent of the occlusion.

Implementation details. The network architecture resembles that of Sec. 3.2, but extends the conditioning, motivated by the inpainting setup in [68]. We apply the pre-trained VAE separately to the masked image $I \odot M$ and the context image I , yielding 2×8 channels, and stack them with the 8D noise image and the unencoded part mask M to obtain the 25-channel input to the diffusion model.

3.4. Part reconstruction

Given a multi-view part image $\hat{J}$, the final step is to reconstruct the part in 3D. Because the part views are now complete and consistent, we can simply use the RM to obtain a predicted reconstruction $\hat{S} = \Psi(\hat{J})$ of the part. We found that the RM does not require fine-tuning to reconstruct object parts, so any good-quality reconstruction model is likely to work well here.

3.5. Training data

To train the various models, we require a dataset of 3D objects consisting of multiple parts. We constructed this dataset from a collection of 140k 3D artist-generated assets, licensed for AI training from a commercial source. Each asset $\mathbf{L}$ is stored as a GLTF scene that generally contains several watertight meshes $(\mathbf{S}^1, \dots, \mathbf{S}^S)$. These usually correspond to parts as created by the original artists, for instance for editing convenience. Example objects from the dataset are shown in Fig. 3.

Multi-view generator data. To train the multi-view generator models Φ , first of all, we have to render the target multi-view images I consisting of 4 views of the full object. Following Instant3D [36], we rendered shaded colors I from the 4 views from the orthogonal azimuths and 20° elevation and arranged them in a 2×2 grid. In the case of *text conditioning*, the training data consists of the pairs $\{(I_n, y_n)\}_{n=1}^N$ of multi-view images and their text captions. Following AssetGen [73], we chose the 10k highest-quality assets and generated their text captions using a CAP3D-like pipeline [48] that used the LLAMA3 model [16]. In the case of *image conditioning*, we use all 140k models, and the conditioning y_n comes in the form of single renders from a randomly sampled direction.

Part segmentation and completion data. To train the part segmentation and completion networks, we need to additionally render the multi-view part images and their depth maps. We filter the dataset to avoid having excessively granular parts that likely lack semantic meaning. To this end, we first cull the parts that take less than 5% of the object volume, and then remove the assets that have more than 10 parts or consist of a single monolithic part. This results in a dataset of 45k objects containing a total of 210k parts. Given an object $\mathbf{L} = (\mathbf{S}_s)_{s=1}^S$, we render the corresponding multi-view images $(J^s)_{s=1}^S$ (shown in Fig. 3) and the corresponding depth maps $(\delta^s)_{s=1}^S$ from the same viewpoints.

The *segmentation diffusion network* is trained on the dataset of pairs $\{(I_n, \mathbf{M}_n)\}_{n=1}^N$, where the segmentation map $\mathbf{M} = [M^k]_{k=1}^S$ is a stack of multi-view binary part masks $M^k \in \{0, 1\}^{2H \times 2W}$. Each mask shows the pixels where the corresponding part is visible in I : $M_{i,j}^k = [k = \text{argmin}_l \delta_{i,j}^l]$, where $k, l \in \{1, \dots, S\}$ and brackets denote Iverson brackets. The *part completion network* is trained on

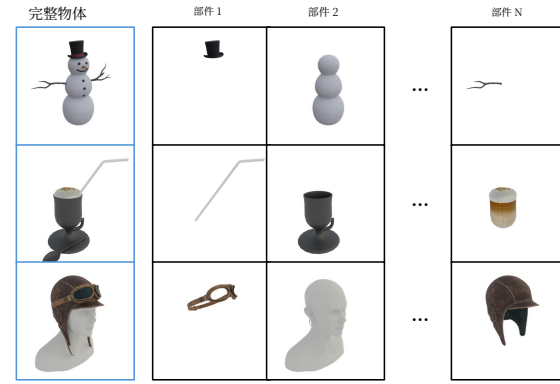


图3。 **训练数据。** 我们从艺术家创建的资产中获取了一个由分解为部件的三维物体组成的数据集。这些物体根据艺术家的设计“自然地”分割成各个部件。

3.3. 上下文部件补全

到目前为止，我们的方法生成了三维物体的多视图图像 I 以及二维分割区域 $M^1, M^2, \dots, M^S$ 。接下来需要将这些结果转换为完整的三维部件重建。给定一个掩码 M ，我们可以将掩码图像 $I \odot M$ 提交给 RM，以获得该部件的三维重建，即， $\hat{\mathbf{S}} = \Psi(I \odot M)$ 。然而，某些部件可能会被其他部件严重遮挡，在极端情况下甚至可能完全不可见。虽然我们可以训练 RM 来处理此类遮挡，但在实践中，这种方法效果并不好，因为部件补全本质上是一个随机问题，而 RM 是确定性的。

为了解决这一问题，我们再次改造多视图生成器，这次用于执行部件补全。后一个模型能够根据文本或单张图像生成完整物体，因此，经过适当微调后，它能够幻觉式地生成部件的任意缺失部分。

形式上，我们将！微调为补全模型！ comp ，该模型将掩码多视图部件图像 $I \odot M$ 映射为补全后的多视图部件图像 $\hat{J} \sim p(\hat{J} \mid \text{! comp}, I \odot M)$ 。然而，我们注意到，有时部件几乎不可见，因此掩码图像 $I \odot M$ 提供的信息非常有限。由于我们需要生成的部件与其他部件及完整物体良好适配，因此我们向模型提供未掩码图像 I 作为上下文。因此，我们基于掩码图像 $I \odot M$ 、未掩码图像 I 和掩码 M 对 $p(\hat{J} \mid I \odot M, I, M)$ 进行条件化。上下文 I 的重要性会随着遮挡程度的增加而提高。

实现细节。 网络架构类似于第 3.2 节中的架构，但受到 [68] 中的图像修复设置启发，我们扩展了条件输入。我们将预训练 VAE 分别应用于掩码图像 $I \odot M$ 和上下文图像 I ，得到 2×8 个通道，并将它们与 8D 噪声图像以及未编码的部件掩码 M 堆叠起来，从而得到扩散模型的 25 通道输入。

3.4. 部件重建

给定一个多视图部件图像 $\hat{J}$ ，最后一步是在三维中重建该部件。由于部件视图现在已经完整且一致，我们可以直接使用 RM 获得该部件的预测重建结果 $\hat{\mathbf{S}} = \Psi(\hat{J})$ 。我们发现，RM 无需微调即可重建物体部件，因此任何高质量的重建模型都可能在此发挥良好作用。

3.5. 训练数据

为了训练各种模型，我们需要一个由包含多个部件的三维物体组成的数据集。我们从 14 万个三维艺术家生成的资产中构建了该数据集，这些资产已获得商业来源授权，可用于人工智能训练。每个资产 $\mathbf{L}$ 都以 GLTF 场景的形式存储，通常包含若干个水密网格 $(\mathbf{S}^1, \dots, \mathbf{S}^S)$ 。这些网格通常对应于原始艺术家创建的部件，例如为了便于编辑。数据集中的物体示例见图 3。

多视图生成器数据。 为了训练多视图生成器模型！，首先，我们必须渲染目标多视角图像 I ，这些图像由完整物体的 4 个视图组成。遵循 Instant3D [36]，我们从正交方位角和 20° 俯仰角的 4 个视图中渲染阴影颜色 I ，并将它们排列成 2×2 网格。对于文本条件，训练数据由多视角图像及其文本描述组成的成对数据 $\{(I_n, y_n)\}_{n=1}^N$ 构成。遵循 AssetGen [73]，我们选取了质量最高的 1 万个资产，并使用类似 CAP3D 的流程 [48] 为其生成文本描述，该流程使用了 LLAMA3 模型 [16]。对于图像条件，我们使用全部 14 万个模型，条件 y_n 的形式是从随机采样方向渲染得到的单张图像。

部件分割与补全数据。 为了训练部件分割与补全网络，我们还需要额外渲染多视图部件图像及其深度图。我们对数据集进行筛选，以避免包含过于细粒度、可能缺乏语义意义的部件。为此，我们首先剔除占物体体积不足 5% 的部件，然后移除包含超过 10 个部件或由单一整体部件组成的资产。最终得到一个包含 4.5 万个物体、总计 21 万个部件的数据集。给定一个物体 $\mathbf{L} = (\mathbf{S}_s)_{s=1}^S$ ，我们渲染相应的多视图图像 $(J^s)_{s=1}^S$ （如图 3 所示）以及相应的深度图 $(\delta^s)_{s=1}^S$ ，视点与前者相同。

分割扩散网络在由成对数据 $\{(I_n, \mathbf{M}_n)\}_{n=1}^N$ 构成的数据集上进行训练，其中分割图 $\mathbf{M} = [M^k]_{k=1}^S$ 是多视图二值部件掩码 $M^k \in \{0, 1\}^{2H \times 2W}$ 的堆叠。每个掩码显示对应部件在 I 中可见的像素： $M_{i,j}^k = [k = \text{argmin}_l \delta_{i,j}^l]$ ，其中 $k, l \in \{1, \dots, S\}$ ，方括号表示 Iverson 括号。部件补全网络在

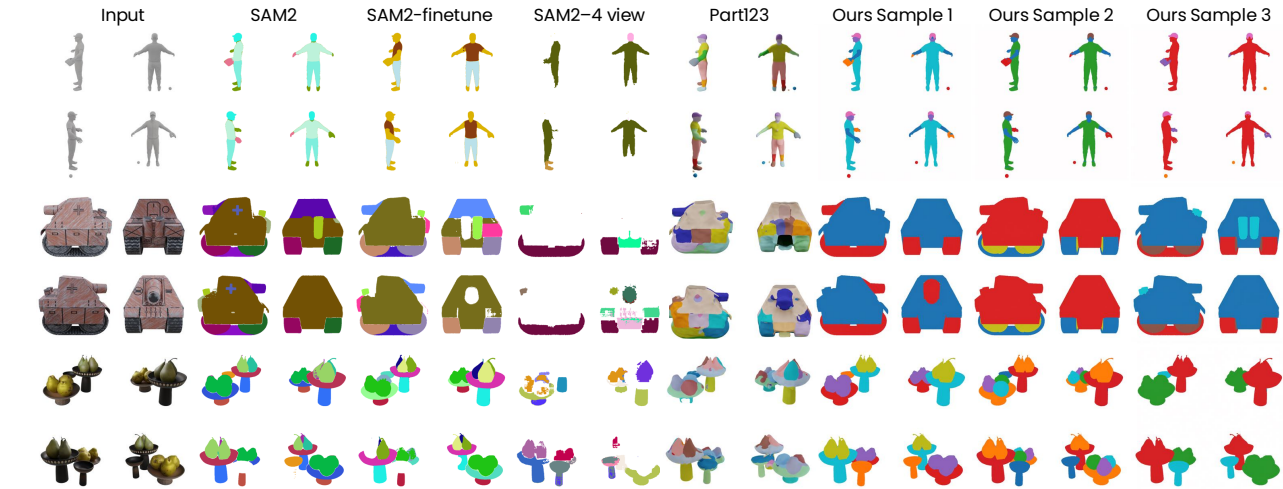


Figure 4. **Examples of automatic multi-view part segmentations.** By running our method several times, we obtain diverse segmentations, covering the space of artist intents better than alternatives.

Method	Automatic		Seeded	
	mAP ₅₀ ↑	mAP ₇₅ ↑	mAP ₅₀ ↑	mAP ₇₅ ↑
Part123 [40]	11.5	7.4	10.3	6.5
SAM2 [†] [67]	20.3	11.8	24.6	13.1
SAM2* [67]	37.4	27.0	44.2	30.1
SAM2 [67]	35.3	23.4	41.4	27.4
PartGen (1 sample)	45.2	32.9	44.9	33.5
PartGen (5 samples)	54.2	33.9	51.3	32.9
PartGen (10 samples)	59.3	38.5	53.7	35.4

Table 1. **Segmentation results.** SAM2* is fine-tuned on our data and SAM2[†] is fine-tuned for multi-view segmentation.

4. Experiments

We first individually evaluate the two main components of our pipeline, namely part segmentation (Sec. 4.1) and part completion and reconstruction (Sec. 4.2). We then evaluate how well the decomposed reconstruction matches the original object (Sec. 4.3). For all experiments, we use 100 held-out objects from the dataset described in Sec. 3.5.

4.1. Part Segmentation

Evaluation protocol. We set up two settings for the segmentation task. One is *automatic part segmentation*, where the input is the multi-view image I , and the method is required to output all parts of the object $M^1, \dots, M^S$. The other is *seeded segmentation*, where we assume that users provide a point as an additional input for a specific mask. In this case, the segmentation algorithm is regarded as a black box $\hat{\mathbf{M}} = \mathcal{A}(I)$ mapping the multi-view image I to a ranked list of N part segmentations (which can, in general, partially overlap). This ranked list is obtained by scoring candidate regions and removing redundant ones — more details are provided in the sup. mat. We then match these segments to the ground-truth segments M_k and report *mean Average Precision* (mAP). This precision can be low due to the inherent ambiguity of the problem: the parts predicted by the algorithm will not match any particular artist’s choice.

Baselines. We consider the original and fine-tuned SAM2 [67] as our baselines for multi-view segmentation. We fine-tune SAM2 in two different ways. First, we fine-tune SAM2’s mask decoder on our dataset, given the ground-truth masks and randomly selected seed points for different views. Second, we concatenate the four orthogonal views in a multi-view image I and fine-tune SAM2 to predict the multi-view mask $\mathbf{M}$ (in this case, the seed point randomly falls in one of the views). SAM2 produces three

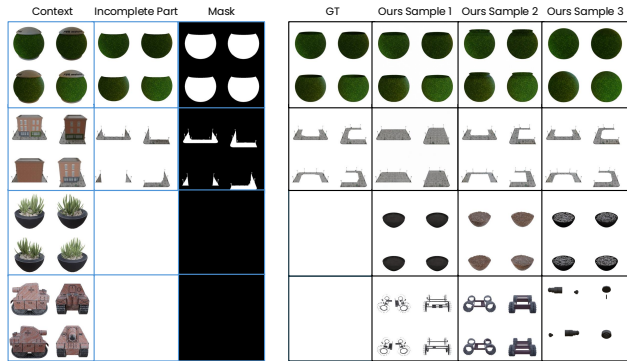


Figure 5. **Qualitative results of part completion.** The images with blue borders are the inputs. Our algorithm produces various plausible completions across different runs. Even when given an empty (*i.e.*, invisible) part, PartGen attempts to generate internal structures inside the object, such as sand or inner wheels.

the dataset of triplets $\{(I_{n'}, J_{n'}, M_{n'})\}_{n'=1}^{N'}$. All the components are produced in the way described above.

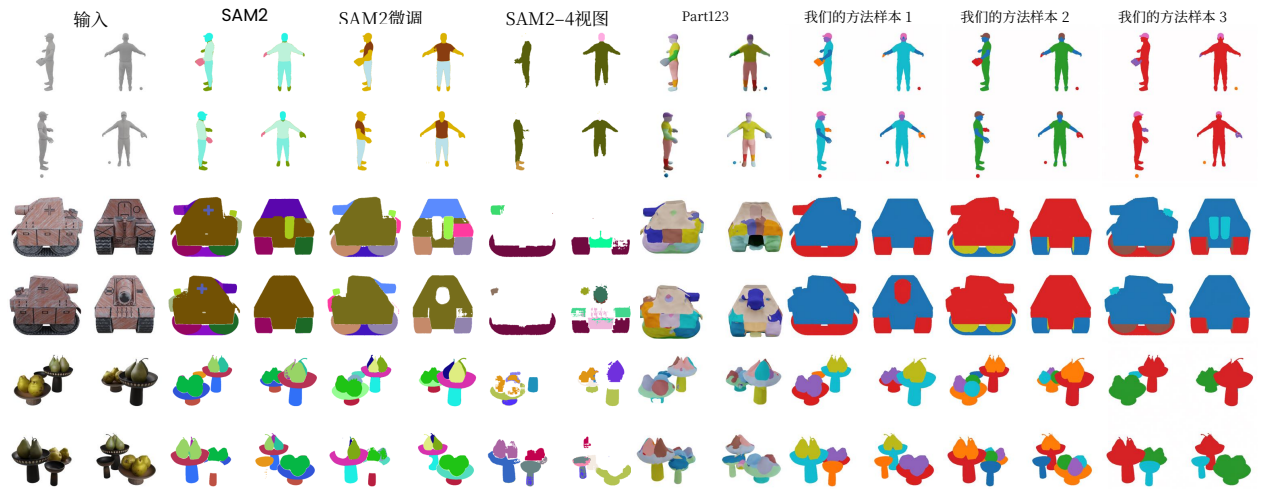


图 4. 自动多视图部件分割示例。通过多次运行我们的方法，可以获得多样化的分割结果，相较于其他方法，能够更好地覆盖艺术家意图的空间。

方法	自动			有种子	
	平均精度均值 ₅₀ ↑	平均精度均值 ₇₅ ↑	平均精度均值 ₅₀ ↑	平均精度均值 ₇₅ ↑	
Part123 [40]	11.5	7.4	10.3		6.5
SAM2 [†] [67]	20.3		11.8	24.6	13.1
SAM2* [67]	37.4		27.0	44.2	30.1
SAM2 [67]	35.3		23.4	41.4	27.4
PartGen (1 个样本)	45.2		32.9	44.9	33.5
PartGen (5 个样本)	54.2		33.9	51.3	32.9
PartGen (10 个样本)	59.3		38.5	53.7	35.4

表1. 分割结果。SAM2* 在我们的数据上进行微调，SAM2[†] 针对多视图分割进行微调。

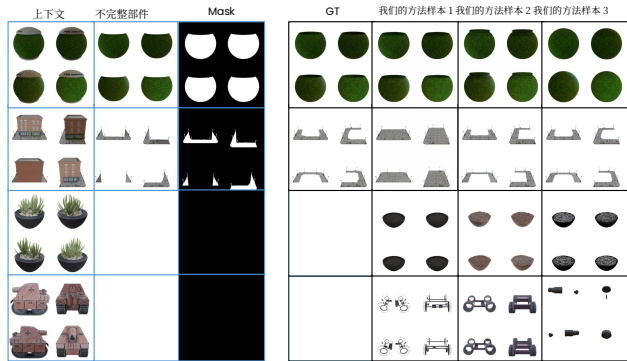


图5. 部件补全的定性结果。带有蓝色边框的图像是输入。我们的算法在不同运行中生成了各种合理的补全结果。即使输入的是空部件（即不可见），PartGen 也会尝试在对象内部生成内部结构，例如沙子或内部车轮。

三元组数据集 $\{(I_{n'}, J_{n'}, M_{n'})\}_{n'=1}^{N'}$ 上进行训练。所有组件均按照上述方式生成。

4. 实验

我们首先分别评估流程的两个主要组成部分，即部件分割（第 4.1 节）以及部件补全与重建（第 4.2 节）。随后，我们评估分解重建与原始对象的匹配程度（第 4.3 节）。对于所有实验，我们使用数据集中描述于第 3.5 节的 100 个留出对象。

4.1. 部件分割

评估协议。 我们为分割任务设置了两种情形。一种是自动部件分割，其中输入是多视图图像 I ，方法需要输出对象的所有部件 $M^1, \dots, M^S$ 。另一种是种子分割，其中我们假设用户提供一个点作为特定掩码的额外输入。在这种情况下，分割算法被视为黑盒 $\hat{\mathbf{M}} = \mathcal{A}(I)$ ，将多视图图像 I 映射为 N 部件分割的排序列表（通常可能部分重叠）。该排序列表通过对候选区域进行评分并移除冗余区域获得——更多细节见补充材料。随后，我们将这些分割区域与真实分割区域 M_k 进行匹配，并报告平均精度均值（mAP）。由于该问题固有的歧义性，这一精度可能较低：算法预测的部件不会与任何特定艺术家的选择完全匹配。

基线方法。 我们将原始 SAM2 [67] 和微调后的 SAM 视为多视图分割的基线方法。我们采用两种不同的方式对 SAM2 进行微调。首先，在给定真实掩码以及为不同视图随机选择的种子点的情况下，在我们的数据集上对 SAM2 的掩码解码器进行微调。其次，将多视图图像中的四个正交视图拼接起来 I ，并对 SAM2 进行微调，使其预测多视图掩码 $\mathbf{M}$ （在这种情况下，种子点会随机落在其中一个视图中）。对于每个输入图像和种子点，SAM2 会生成三个

Method	Compl.	Multi-view	Context	View completion $\hat{J}$			3D reconstruction $\hat{S}$		
				CLIP↑	LPIPS↓	PSNR↑	CLIP↑	LPIPS↓	PSNR↑
Oracle ($\hat{J} = J$)	GT	—	—	1.0	0.0	∞	0.957	0.027	18.91
PartGen ($\hat{J} = \mathcal{B}(I \odot M, I)$)	✓	✓	✓	0.974	0.015	21.38	0.936	0.039	17.16
w/o context ($\hat{J} = \mathcal{B}(I \odot M)$)	✓	✓	✗	0.951	0.028	16.80	0.923	0.046	14.83
single view ($\hat{J}_v = \mathcal{B}(I_v \odot M_v, I_v)$)	✓	✗	✓	0.944	0.031	15.92	0.922	0.051	13.25
None ($\hat{J} = I \odot M$)	✗	—	—	0.932	0.039	13.24	0.913	0.059	12.32

Table 2. **Part completion results.** We first evaluate view part completion by computing scores w.r.t. the ground-truth multi-view part image J . Then, we evaluate 3D part reconstruction by reconstructing each part S and rendering it. See text for details.

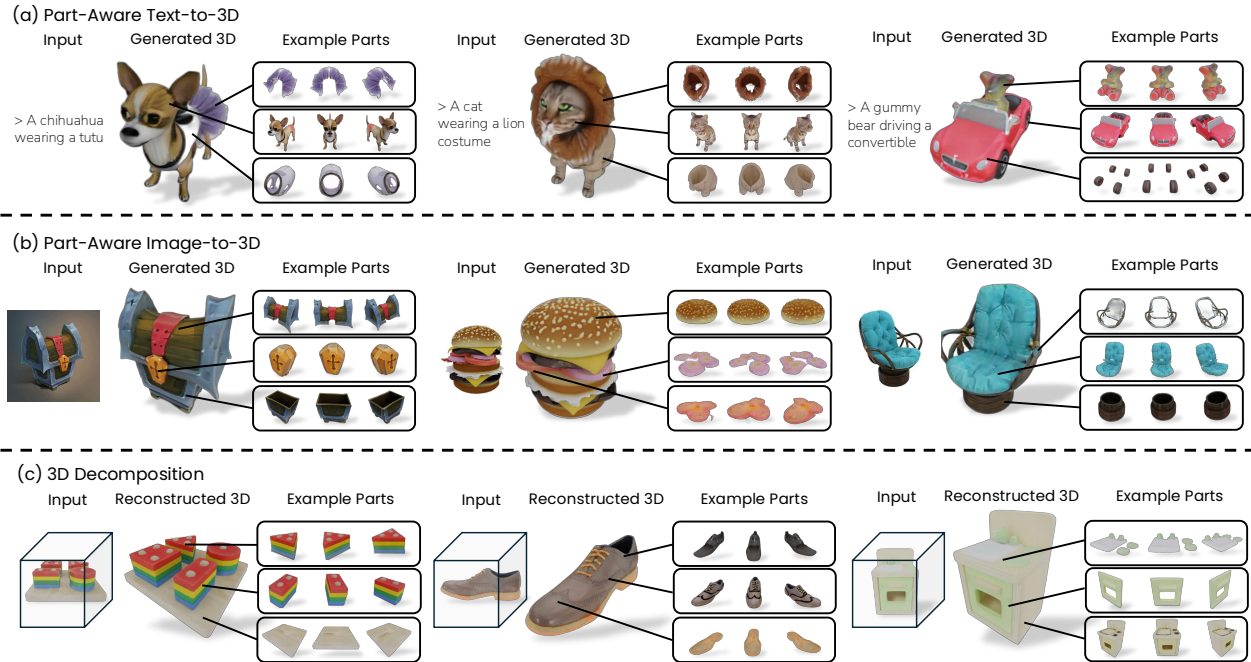


Figure 6. **Examples of applications.** PartGen can effectively generate or reconstruct 3D objects with meaningful and realistic parts in different scenarios: a) Part-aware text-to-3D generation; b) Part-aware image-to-3D generation; c) 3D decomposition. Some assets, like the hamburger and the blocks, consist of more than three parts, but we display only three for a more visually balanced arrangement.

regions for each input image and seed point. For automatic segmentation, we seed SAM2 with a set of query points spread over the object, obtaining three different regions for each seed point. For seeded segmentation, we simply return the regions that SAM2 outputs for the given seed point. We also provide a comparison with recent work, Part123 [40].

Results. We report the results in Tab. 1. As shown in the table, mAP results for our method are *much* higher than others, including SAM2 fine-tuned on our data. This is primarily because of the ambiguity of the segmentation task, which is better captured by our generator-based approach. We further provide qualitative comparisons in Fig. 4.

4.2. Part completion and reconstruction

We utilize the same test data as in Sec. 4.1, forming tuples (S^k, I, M^k, J^k) consisting of the 3D object part S^k , the full multi-view image I , the part mask M^k , and the multi-view

image J^k of the part, as described in Sec. 3.5. We choose one random part index k per model and omit it from the notation below for conciseness.

Evaluation protocol. The completion algorithm and its baselines are treated as a black box $\hat{J} = \mathcal{B}(I \odot M, I)$ that predicts the completed multi-view image $\hat{J}$. We then compare $\hat{J}$ to the ground-truth render J using Peak Signal-to-Noise Ratio (PSNR) of the foreground pixels, Learned Perceptual Image Patch Similarity (LPIPS) [97], and CLIP similarity [66]. The latter is an important metric since the completion task is highly ambiguous, which motivates evaluating *semantic* similarity. We also evaluate the quality of the reconstruction of the predicted completions by comparing the reconstructed object part $\hat{S} = \Psi(\hat{J})$ to the ground-truth part S using the same metrics, but averaged after rendering the part to four random novel viewpoints.

(a) 三维分解
输入 重建的三维模型 示例部件 输入 重建的三维模型 示例部件 输入 重建的三维模型 示例部件

方法	视图补全 $\hat{J}$					三维重建 $\hat{S}$		
	补全多视图上下文	CLIP↑	LPIPS↓	PSNR↑	CLIP↑	LPIPS↓	PSNR↑	
预言机 ($\hat{J} = J$)	GT	—	—	1.0	0.0	∞	0.957	0.027 18.91
PartGen ($\hat{J} = \mathcal{B}(I \odot M, I)$)	✓	✓	✓	0.974	0.015	21.38	0.936	0.039 17.16
无上下文 ($\hat{J} = \mathcal{B}(I \odot M)$)	✓	✓	✗	0.951	0.028	16.80	0.923	0.046 14.83
单视图 ($\hat{J}_v = \mathcal{B}(I_v \odot M_v, I_v)$)	✓	✗	✓	0.944	0.031	15.92	0.922	0.051 13.25
无 ($\hat{J} = I \odot M$)	✗	—	—	0.932	0.039	13.24	0.913	0.059 12.32

表2. **部件补全结果。**我们首先通过计算相对于真实多视图部件图像 J 的得分来评估视图部件补全。然后，我们通过重建每个部件 S 并对其进行渲染，来评估三维部件重建。详情请参阅正文。

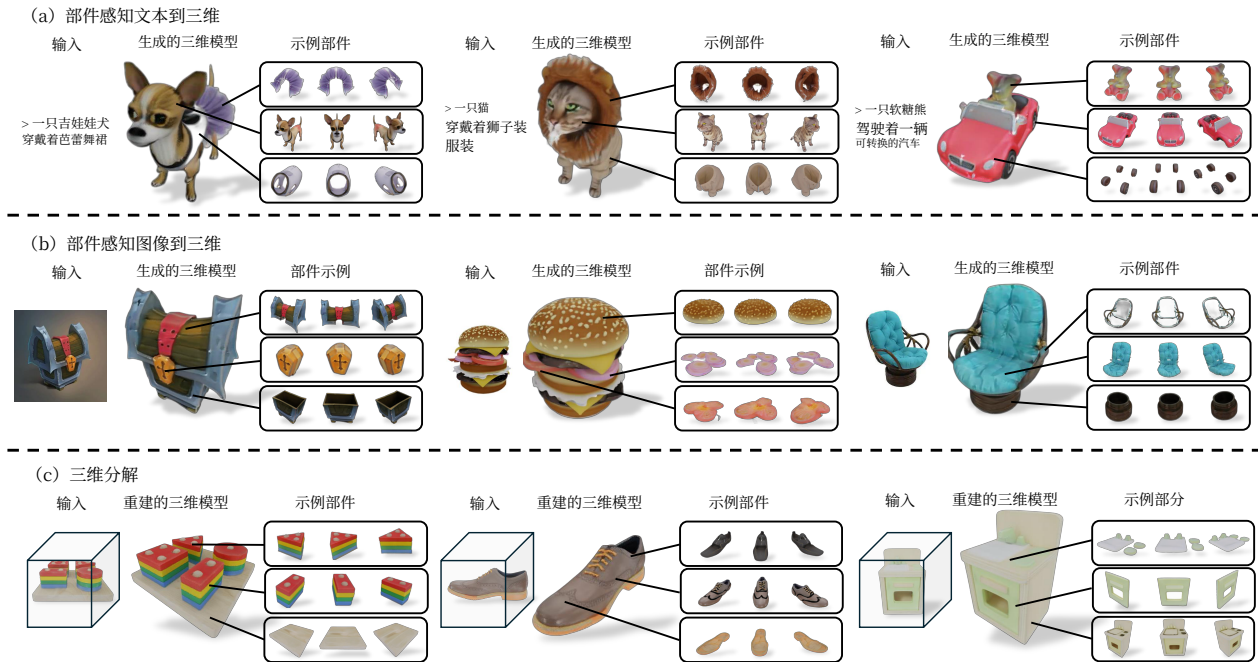


图6. **应用示例。**PartGen能够在不同场景下有效生成或重建具有明确且逼真部件的三维物体：a) 部件感知文本到三维生成；b) 部件感知图像到三维生成；c) 三维分解。一些资产（如汉堡包和积木）由三个以上的部件组成，但为了实现视觉上更加均衡的排列，我们仅展示其中三个部件。

区域。对于自动分割，我们使用一组分布在物体上的查询点为 SAM2 提供种子，从而为每个种子点获得三个不同的区域。对于有种子分割，我们只返回 SAM2 针对给定种子点输出的区域。我们还将其与近期工作 Part123 [40]进行比较。

结果。我们在表1中报告了结果。如表中所示，我们方法的平均精度均值结果明显高于其他方法，包括在我们的数据上微调后的SAM2。这主要是因为分割任务存在歧义，而我们基于生成器的方法能够更好地捕捉这种歧义。我们进一步在图4中提供了定性比较。

4.2. 部件补全与重建

我们使用与第4.1节相同的测试数据，构成元组 (S^k, I, M^k, J^k) ，其由三维物体部件 S^k 、完整多视图图像 I 、部件掩码 M^k 和多视图

部件的图像 J^k 组成，如第3.5节所述。我们为每个模型随机选择一个部件索引 k ，为简洁起见，以下记法中省略该索引。

评估协议。我们将补全算法及其基线方法视为黑盒 $\hat{J} = \mathcal{B}(I \odot M, I)$ ，该黑盒会预测补全后的多视图图像 $\hat{J}$ 。随后，我们使用前景像素的峰值信噪比（PSNR）、学习感知图像块相似度（LPIPS）[97]，和 CLIP相似度 [66]，将 $\hat{J}$ 与真实渲染图 J 进行比较。后者是一项重要指标，因为补全任务具有高度歧义性，这促使我们评估语义相似度。我们还通过将重建的物体部件 $\hat{S} = \Psi(\hat{J})$ 与真实部件 S 进行比较，评估预测的补全结果的重建质量；具体而言，在将该部件渲染到四个随机新视点后，使用相同的指标取平均值。



Figure 7. **3D part editing.** We can edit the appearance and shape of the 3D objects with text prompt.

Method	CLIP↑	LPIPS↓	PSNR↑
PartGen ($\hat{\mathbf{L}} = \bigcup_k \Psi(\hat{J}_k)$)	0.952	0.065	20.33
Unstructured ($\hat{\mathbf{L}} = \Psi(I)$)	0.955	0.064	20.47

Table 3. **Model reassembling results.** The quality of 3D reconstruction of the object as a whole is close to that of the part-based one, showing that the predicted parts fit together well.

Results. We compare our part completion algorithm ($\hat{J} = \mathcal{B}(I \odot M, I)$) to several baselines: using no completion ($\hat{J} = I \odot M$), omitting the context ($\hat{J} = \mathcal{B}(I \odot M)$), and completing single views independently ($\hat{J}_v = \mathcal{B}(I_v \odot M_v, I_v)$). The latter two are fine-tuned based on Φ . We also compare to an oracle ($\hat{J} = J$), that uses complete ground-truth parts, providing the upper bound on part reconstruction performance, where the only bottleneck is the RM.

As shown in Tab. 2, our model largely surpasses the baselines. Both joint multi-view reasoning and contextual part completion are important for good performance. We further provide qualitative results in Fig. 5.

4.3. Reassembling parts

Evaluation protocol. Starting from the multi-view image I of a 3D object $\mathbf{L}$, we run the segmentation algorithm to obtain segmentation $(\hat{M}^1, \dots, \hat{M}^S)$, reconstruct each 3D part as $\hat{\mathbf{S}}^k = \Psi(\hat{J}^k)$, and reassemble the 3D object $\hat{\mathbf{L}}$ by merging the 3D parts $\{\hat{\mathbf{S}}^1, \dots, \hat{\mathbf{S}}^N\}$. Detailed implementation of part reassembling is included in the sup. mat. We then compare $\hat{\mathbf{L}} = \bigcup_k \Psi(\hat{J}_k)$ to the whole-object reconstruction $\hat{\mathbf{L}} = \Psi(I)$, *i.e.* without decomposing the object into parts, using the same protocol as for our method.

Results. Table 3 shows that our method achieves performance comparable to directly reconstructing the objects using the RM ($\hat{\mathbf{L}} = \Psi(I)$), with the additional benefit of producing the reconstruction structured into parts, which are useful for downstream applications such as editing.

4.4. Applications

Part-aware text-to-3D generation. First, we apply PartGen to part-aware text-to-3D generation. We train a text-to-multi-view generator similar to [73], which takes a text prompt as input and outputs a grid of four views. For illustration, we use the prompts from DreamFusion [62]. As shown in Fig. 6, PartGen can effectively generate 3D objects with distinct and completed parts, even in challenging cases with heavy occlusions, such as the gummy bear. Additional examples are provided in the sup. mat.

Part-aware image-to-3D generation. Next, we consider part-aware image-to-3D generation using images from [23, 86]. Building upon the text-to-multi-view generator, we further fine-tune the generator to accept images as input with a strategy similar to [92]. Further training details are provided in the sup. mat. Results are shown in Fig. 6, demonstrating that PartGen is successful in this case too.

Real-world 3D object decomposition. PartGen can also decompose real-world 3D objects. We demonstrate this using objects from Google Scanned Objects (GSO) [15]. Given a 3D object from GSO, we render different views to obtain an image grid and then apply PartGen as described above. The last row of Fig. 6 shows that PartGen can effectively decompose real-world 3D objects.

3D part editing. Finally, we show that once the 3D parts are decomposed, they can be further modified individually. As illustrated in Fig. 7, a variant of our method enables effective editing of the shape and texture of parts based on textual prompts. This part-based editing prevents unintended modifications to other parts of the 3D object. The details of the 3D editing model are provided in the sup. mat.

5. Conclusion

We have introduced PartGen, an approach to generate 3D objects as compositions of 3D parts, given as input text, images, or unstructured 3D objects. PartGen can reconstruct 3D parts that are minimally or even not visible, utilizing the guidance of a multi-view diffusion prior. We have also shown several applications of PartGen, including text-guided part editing. This is a promising step toward the generation of 3D assets that are more useful in professional workflows.

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图7. **三维部件编辑.** 我们可以通过文本提示编辑三维物体的外观和形状。

方法	CLIP↑	LPIPS↓	PSNR↑
PartGen ($\hat{\mathbf{L}} = \bigcup_k \Psi(\hat{J}_k)$)	0.952	0.065	20.33
Unstructured ($\hat{\mathbf{L}} = \Psi(I)$)	0.955	0.064	20.47

表 3. **模型重组结果.** 整体对象的三维重建质量接近基于部件的重建结果，表明预测出的部件能够很好地拼接在一起。

结果. 我们将我们的部件补全算法 ($\hat{J} = \mathcal{B}(I \odot M, I)$) 与几种基线方法进行比较：不进行补全 ($\hat{J} = I \odot M$)、省略上下文 ($\hat{J} = \mathcal{B}(I \odot M)$)，以及独立补全单视图 ($\hat{J}_v = \mathcal{B}(I_v \odot M_v, I_v)$)。后两种方法基于!进行微调。我们还与一个预言机 ($\hat{J} = J$) 进行比较，该预言机使用完整的真实部件，为部件重建性能提供上限，此时唯一的瓶颈是RM。

如表2所示，我们的模型在很大程度上超越了基线方法。联合多视图推理和上下文部件补全对于取得良好性能都很重要。我们还在图5中提供了定性结果。

4.3. 重组部件

评估协议. 从三维物体的多视图图像 I 出发，我们运行分割算法以获得分割结果 $(\hat{M}^1, \dots, \hat{M}^S)$ ，将每个三维部件重建为 $\hat{\mathbf{S}}^k = \Psi(\hat{J}^k)$ ，并通过合并三维部件 $\{\hat{\mathbf{S}}^1, \dots, \hat{\mathbf{S}}^N\}$ 重新组装三维物体 $\hat{\mathbf{L}}$ 。部件重新组装的详细实现见补充材料。随后，我们将 $\hat{\mathbf{L}} = \bigcup_k \Psi(\hat{J}_k)$ 与整体物体重建 $\hat{\mathbf{L}} = \Psi(I)$ 进行比较，即不将物体分解为部件，而是使用与我们的方法相同的协议进行重建。

结果. 表 3 显示，我们的方法使用 RM ($\hat{\mathbf{L}} = \Psi(I)$) 直接重建物体时取得了可比的性能，同时还能够生成按部件组织的重建结果，这些部件可用于编辑等下游应用。

4.4. 应用

部件感知文本到三维生成. 首先，我们将 PartGen 应用于部件感知文本到三维生成。我们训练了一个类似于 [73] 的文本到多视图生成器，该生成器以文本提示作为输入，并输出由四个视图组成的网格。为便于说明，我们使用了 DreamFusion 中的提示词 [62]。如图 6 所示，即使在软糖熊这类存在严重遮挡的挑战性案例中，PartGen 也能有效生成具有明显且补全部件的三维物体。更多示例见补充材料。

部件感知图像到三维生成. 接下来，我们使用来自 [23, 86] 的图像，研究部件感知图像到三维生成。在文本到多视图生成器的基础上，我们进一步对生成器进行微调，使其能够以图像作为输入，所采用的策略类似于 [92]。更多训练细节见补充材料。结果如图 6 所示，表明 PartGen 在该任务中同样表现出色。

真实世界三维对象分解. PartGen 还可以分解真实世界三维物体。我们使用 Google Scanned Objects (GSO) 中的物体对此进行了展示 [15]。给定一个来自 GSO 的三维物体，我们渲染出不同视角以获得图像网格，然后应用上述 PartGen 方法。图 6 的最后一行展示了 PartGen 能够有效地分解真实世界三维物体。

三维部件编辑. 最后，我们展示了三维部件完成分解后，还可以进一步对其进行单独修改。如图 7 所示，我们方法的一个变体能够根据文本提示有效地编辑部件的形状和纹理。这种基于部件的编辑可以避免对三维物体的其他部件进行非预期的修改。三维编辑模型的详细信息见补充材料。

5. 结论

我们提出了 PartGen，这是一种将三维物体生成为三维部件组合的方法，其输入可以是文本、图像或非结构化三维物体。PartGen 利用多视图扩散先验的引导，能够重建最小程度可见甚至完全不可见的三维部件。我们还展示了 PartGen 的多种应用，包括文本引导的部件编辑。这是朝着生成更适用于专业工作流程的三维资产迈出的有意义一步。

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